# LAB 7

**Objective:**

1. WAP to find the sum of 10 numbers in an array.
2. Write a program to find the smallest number in an array of data.
3. Write a program to find the largest number in an array of data.
4. Write a program to generate Fibonacci series up to $10^{th}$ term.

**Program to find sum of 10 numbers in array.**

| Address | Hex Code (Opcode) | Mnemonics | Comments |
|---|---|---|---|
| C000 | 21 | LXI H, C050 | Load 16-bit data C050H to HL pair |
| C001 | 50 | | |
| C002 | C0 | | |
| C003 | 0E | MVI C, 0A | Loads the register C with immediate data 0A H |
| C004 | 0A | | |
| C005 | 3E | MVI A, 00 | Initialize accumulator with 00 |
| C006 | 00 | | |
| C007 | 46 | X: MOV B, M | Move contents of memory to register B |
| C008 | 80 | ADD B | Add the content of register B with accumulator |
| C009 | 23 | INX H | Increment the register pair H-L by 1 |
| C00A | 0D | DCR C | Decrement the register C by 1 |
| C00B | C2 | JNZ X | Jump if no zero to X |
| C00C | 07 | | |
| C00D | C0 | | |
| C00E | 32 | STA CO60 | Store the content of accumulator to memory location C060 |
| C00F | 60 | | |
| C010 | C0 | | |
| C011 | 76 | HLT | Terminate the program. |

**Input:**

Memory Location Data

C050 = 00H

C051 = 01H

C052 = 02H

C053 = 03H

C054 = 04H

C055 = 05H

C056 = 06H

C057 = 07H

C058 = 08H

**Output:**

Memory Location Data

C060 = 2D

C059 =09H

## Program to find the smallest number in an array of data.

| Address | Hex Code (Opcode) | Mnemonics | Comments |
|---|---|---|---|
| C000 | 21 | LXI H, C000 | Load 16-bit data C000H to HL pair |
| C001 | 00 | | |
| C002 | C0 | | |
| C003 | 46 | MOV B, M | Move contents of memory to register B |
| C004 | 23 | INX H | Increment the register pair H-L by 1 |
| C005 | 7E | MOV A, M | Move contents of memory to A |
| C006 | 05 | DCR B | Decrement the register B by 1 |
| C007 | 23 | X: INX H | Increment the register pair H-L by 1 |
| C008 | BE | CMP M | The contents of the Memory is compared with the contents of the accumulator and both contents are preserved. |
| C009 | DA | JC Y | Jump if carry to Y |
| C00A | 0D | | |
| C00B | C0 | | |
| C00C | 7E | MOV A, M | Move contents of memory to A |
| C00D | 05 | Y: DCR B | Decrement the register B by 1 |
| C00E | C2 | JNZ X | Jump if no zero to X |
| C00F | 07 | | |
| C010 | C0 | | |
| C011 | 32 | STA C060 | Store the content of accumulator to memory location C060 |
| C012 | 60 | | |
| C013 | C0 | | |
| C014 | 76 | HLT | Terminate the program. |

**Input:**

Memory Location Data

C000 = 05H

C001 = 02H

C002 = 09H

C003 = 0AH

C004 = 04H

C005 = 08H

**Output:**

Memory Location Data

C060 = 02H

**Program to find the largest number in an array of data.**

## Code:

```
LXI H, C000
        MOV B, M
        INX H
        MOV A, M
        DCR B
X:  INX H
        CMP M
        JNC AHEAD
        MOV A, M
Y: DCR B
        JNZ LOOP
        STA C060
        HLT
```

**Input:**

Memory Location Data

C000 = 05H

C001 = 02H

C002 = 09H

C003 = 0AH

C004 = 04H

C005 = 08H

**Output:**

Memory Location Data

C060 = 0AH

## Program to generate Fibonacci series up to 10th term.

**Code:**

```
LXI H, C050H
        MVI C,08H
        MVI B,00H
        MVI D,01H
        MOV M, B
        INX H
        MOV M, D
X: MOV A, B
        ADD D
        MOV B, D
        MOV D, A
        INX H
        MOV M, A
        DCR C
        JNZ X
        HLT
```

**Output**

| | |
|---|---|
| C051 | 01 |
| C052 | 01 |
| C053 | 02 |
| C054 | 03 |
| C055 | 05 |
| C056 | 08 |
| C057 | 0D |
| C058 | 15 |
| C059 | 22 |